Qualitative Best Practices: Steps 3–9

Contents

Step 3 Open code with in vivo labels
Step 4 Create condensed case timelines
Step 5 Compare cases for second order themes
Step 6 Develop categories and classify cases
Step 7 Iterate between data coding theory and literature
Step 8 Build the findings narrative and evidence display
Step 9 Construct the process model

Step 3 Open code with in vivo labels

A. Qualitative best practices *(from the master document, revised)*

- **Open to the data** | Start with no codebook and let labels come from the material, within the study's focus. | "*This is often called "open coding" (Strauss and Corbin 1998:223) to emphasize the importance of being open to the data.*" (Patton, 2002, pp. 453–454)
- **Use members' own words** | Label codes with participants' own phrases wherever a phrase captures the dynamic. | "*In vivo code: Concepts using the actual words of research participants*" (Corbin & Strauss, 2015, p. 85) Also see (Patton, 2002, pp. 454–455; Gioia et al., 2013, p. 26).
- **Let codes multiply** | Don't collapse or abstract codes yet; a long, messy list is expected at this stage. | "*In this 1st-order analysis, which tries to adhere faithfully to informant terms, we make little attempt to distill categories, so the number of categories tends to explode on the front end of a study.*" (Gioia et al., 2013, p. 20)
- **Several codes per passage** | A segment can carry more than one code. Code by distinct idea, not by unit; zero is acceptable. A sentence or short speaking turn should have no more than 3 codes, and a full paragraph no more than 5. These are ceilings, not targets. Don't assume any number of codes is average, normal or recommended; give each segment as many codes as the data and the research question warrant. | "*Many passages will illustrate more than one theme or pattern.*" (Patton, 2002, p. 463)
- **Code independently, then compare** | Each coder builds codes separately before comparing; disagreements are information, not noise. | "*Where more than one person is working on the analysis, it is helpful to have each person (or small teams for large projects) develop the coding scheme independently, then compare and discuss similarities and differences.*" (Patton, 2002, p. 464) Also see (Eisenhardt, 1989, p. 538).
- **Explicit merge rules** | Resolve differences with written rules for how terms are coded, and apply them consistently. | "*We reconcile differing interpretations by developing consensual decision rules about how various terms or phases are to be coded.*" (Gioia et al., 2013, p. 22) ("phases" as in the original)
- **Delay the literature** | Keep coders away from any specific theory at this stage to avoid any confirmation bias. | "*Up to this stage in the research, we make a point of not knowing the literature in great detail, because knowing the literature intimately too early puts blinders on and leads to prior hypothesis bias (confirmation bias).*" (Gioia et al., 2013, p. 21)

B. Further reading

B1. Corbin, J., & Strauss, A. (2015). Basics of Qualitative Research (4th ed.). Chapter 12, "Open Coding: Identifying Concepts," pp. 220–238.

The master document lists this reading as "Saldana, 2025"; the chapter is Corbin & Strauss (2015). The project copy's page numbers don't match the print edition, so quotations from it are cited by chapter rather than page.

The chapter is a worked demonstration rather than a list of rules. The author open-codes the first interview of a study of war veterans, writing a memo for each piece of data and pausing for methodological notes. Table 12.1 sets the vocabulary. Coding is "Delineating concepts to stand for interpreted meaning of data." Concepts are "Words that stand for interpreted meaning of data," which let the analyst "group 'raw data' with other 'raw data' that share a common

meaning or characteristic." Categories are "Higher-level concepts under which analysts group lower-level concepts." Properties describe a concept, and dimensions are "Variations within properties."

Before coding. The review that opens the demonstration (Part 2 introduction) sets out the preparatory moves. Read the whole document first, "resisting the urge to write in the margins," so you hold "a mental picture of the whole interview or observation." Then "look for the natural breaks in the document," making "some arbitrary breaks" where a speaker runs on. For each section, either step back and ask "What in the larger sense seems to be going on in this section of data?" or go line by line, which "is especially useful if data are not clear." Most analysts combine the two. Once the main idea is clear, ask "What is the action being taken? Who is doing it? When? Where? How? Why?" and give each answer a conceptual name.

Two levels of concept. In Memo 1, a paragraph of background talk gets one main concept, "locating" (the self at entry to the military), and lower-level concepts ("family background," "patriotism," "path to war") that "are the qualifiers or extenders of 'locating.'" The author warns: "One of the mistakes made by beginning analysts is the failure to differentiate between levels of concepts. They just put labels on concepts and don't start early in the analytic process building concepts." Each memo "has been assigned a number, is dated, and is titled with a concept," and the analyst "always wants to come up with the most abstract concept possible so that a concept has application beyond this one incident or participant."

Code the essence, often in the speaker's words. Labels should capture what is meant, not the surface topic: "I could have coded the interviewee's feelings as 'change in attitude' or as 'losing patriotism' as beginning researchers often do. But what an analyst tries to do is to get at the essence of what is being said." Participants' phrases enter the analysis when they carry that essence: "nips at [his] conscience," "hardened," and "Hey, this is war!" are all used in the speaker's own words.

Comparisons, memos, and provisional codes. When the interviewee turned out to be a noncombatant, the author made a "theoretical comparison" with combatants from general knowledge, but "None of the ideas on 'combatants' that came out of 'theoretical comparison' will be put into the study." The comparison only sharpened the meaning of the data and pointed to who to interview next. Everything stays open: "At the beginning of analysis, all concepts and any categories or themes are considered provisional and will be checked out against further data and either added to, discarded, or modified." The first interview yielded 15 memos, a list of 24 concepts, and three tentative themes that "remain unverified and undeveloped." The list is not the point: "just listing concepts is not what doing grounded theory is about." On a team, members discuss the data together, and the author recommends that "one person be the designated 'note keeper.'"

B2. Patton, M. Q. (2002). Qualitative Research and Evaluation Methods (3rd ed.). "Coding Data, Finding Patterns, Labeling Themes, and Developing Category Systems" and "Convergence and Divergence in Coding and Classifying," pp. 462–467.

Patton treats coding as the first act of content analysis: "Developing some manageable classification or coding scheme is the first step of analysis. Without classification there is chaos and confusion" (p. 463). He describes the work without software, because "the principles of the analytical process are the same whether doing it manually or with the assistance of a computer program" (p. 463).

Mechanics. The first cut is a read-through with notes in the margins: "Coming up with topics is like constructing an index for a book or labels for a file system: You look at what is there and

give it a name, a label" (p. 463). Exhibit 8.7 shows such first-cut codes from a program evaluation: short abbreviations written on the passage, each standing for a file of all similarly coded passages, such as "Ps Re Prog" (participants' reactions to the program) or "C-SP" (conflicts between staff and participants) (p. 464). Coding takes several passes: "The first reading through the data is aimed at developing the coding categories or classification system. Then a new reading is done to actually start the formal coding in a systematic way. Several readings of the data may be necessary before field notes or interviews can be completely indexed and coded" (p. 463).

Several coders. When more than one person analyzes, each builds the scheme independently and then compares, because "Important insights can emerge from the different ways in which two people look at the same set of data, a form of analytical triangulation" (p. 464). In a study of 60 interviews, "every paragraph in every interview was coded using a systematic and comprehensive coding scheme made up of 15 general categories with subcategories," and "Every interview was coded twice by two independent coders" (pp. 464–465). Every code was stored so that the analyst could go back to the full passage and interview "to keep quotations in context" (p. 465). Patton notes that such labor-intensive systems suit large data sets used by several people over time (p. 465).

Convergence and divergence. As codes accumulate, the analyst looks for "recurring regularities" and sorts them into categories, which "should then be judged by two criteria: internal homogeneity and external heterogeneity" (p. 465). Quoting Guba: "The existence of a large number of unassignable or overlapping data items is good evidence of some basic fault in the category system" (p. 465). A finished set of categories should hang together as "a whole picture," be "reasonably inclusive of the data," be "reproducible by another competent judge," and be "credible to the persons who provided the information" (p. 466). The mirror move, divergence, fleshes categories out by extension, bridging and surfacing, and includes "careful and thoughtful examination of data that doesn't seem to fit" (p. 466). Patton adds that this sequence "should not be followed mechanically, linearly, or rigidly" (p. 466), and that qualitative findings are judged "by their substantive significance" rather than statistical significance (p. 467).

Step 4 Create condensed case timelines

A. Qualitative best practices *(from the master document, revised)*

- **Analytic chronology** | The timeline should line up events with early themes, not just list what happened. | "*Our analytical chronologies reach towards theory presentation but are prepared to get on top of the data, to clarify sequences across levels of analysis, suggest causal linkages between levels, and establish early analytical themes.*" (Pettigrew, 1990, p. 280) Also see (Yin, 2003, p. 125).
- **Events and codes in parallel** | Show events, codes and sources as parallel tracks on one time axis. | "*Visual graphical representations are particularly attractive for the analysis of process data because they allow the simultaneous representation of a large number of dimensions, and they can easily be used to show precedence, parallel processes, and the passage of time.*" (Langley, 1999, p. 700) Also see (Eisenhardt, 1989, p. 540).
- **Define what counts as an event** | An event is something that happened, was done or was decided and is pertinent to the research question. Decide in advance which kinds and levels of events go on the timeline and apply that choice to every case. Give events that unfold over time a start and an end date, and record gradual trends and inner states in notes rather than forcing them into events. | "*Despite the apparent temporal precision indicated by the word "event," there are also clearly different levels of events: an event may include a bad year, a merger, a decision, a meeting, a conversation, or a handshake.*" (Langley, 1999, p. 693)
- **Bracket into periods, not stages** | Split each case's timeline at its discontinuities, points where the pattern of activity changes markedly, without assuming a developmental sequence. | "*Many temporal processes can be decomposed in this way, at least partly, without presuming any progressive developmental logic.*" (Langley, 1999, p. 703) Also see (Yin, 2003, p. 126).
- **Corroborate with documents** | Check event dates from interviews against independent records, and follow up contradictions rather than resolving them silently. | "*For case studies, the most important use of documents is to corroborate and augment evidence from other sources.*" (Yin, 2003, p. 87) Also see (Patton, 2002, p. 559).

B. Further reading

B1. Langley, A. (1999). Strategies for theorizing from process data. Academy of Management Review, 24(4). "Visual Mapping Strategy," pp. 700–703.

Langley presents visual mapping as one of several strategies for making sense of process data. Matrix and graphical forms "allow the presentation of large quantities of information in relatively little space, and they can be useful tools for the development and verification of theoretical ideas" (p. 700).

The worked example. Figure 2, "Extract from a Process Flowchart" (from Langley & Truax, 1994), shows one firm's adoption of new technology as "an event chronology coded in multiple ways" (p. 700). Each event is a box. The box's shape shows the kind of event: "a decision (round-cornered rectangles), an activity (sharp-cornered rectangles), or an event outside the control of the firm (ovals)" (p. 700). Its position in one of six horizontal bands shows the issue domain (product/market, finance, technology, human resources, operations, other activities), and boxes that cross several bands mark integrative events. Arrows to the central band code each event's effect on the focal process: "positive effect [+], negative effect [-], precipitating

effect [++], reorienting effect [0]" (p. 702). Line thickness shows continuity between linked events, and the time scale "allows representation of event ordering and parallel tracks over time and provides a rough indication of their temporal duration" (p. 702). Crucially, "the link between it and the qualitative database is maintained through the use of short descriptions of each element in its corresponding box" (p. 702).

What the drawing is for. "This type of drawing obviously is not a 'theory' but an intermediary step between the raw data and a more abstract conceptualization" (p. 702). The next moves are to "compare several such representations to look for common sequences of events and common progressions in sources of influence," and then to build local causal maps (p. 702). Because the approach requires "many observations of similar processes," it "may be most fruitful as a theory development tool for the analysis of multiple holistic or embedded cases" (p. 702).

Limits. Graphs keep some ambiguity but exclude other things: Figure 2 "gives no room to such factors as power, conflict, and emotion" (p. 702). Precedence, authority and influence are easy to draw, but "emotions and cognitions are less easy to express in this way, being more difficult to temporally pin down" (p. 702). Used alone, the method's conclusions "can have a rather mechanical quality, dealing more with the surface structure of activity sequences than with the underlying forces driving them" (pp. 702–703).

B2. Pettigrew, A. M. (1990). Longitudinal field research on change: Theory and practice. Organization Science, 1(3). "(4) Research Outputs, Audience, and Presentation," pp. 279–281.

Pettigrew places the timeline within a sequence of research outputs. Comparative case research is iterative and "at times untidy" (p. 279), so researchers need to be clear about "the varieties of research output achievable within their mode of operation; the sequencing with which those outputs are deliverable over time; the audiences for which the outputs are intended, and the most suitable form of presentation for each audience" (p. 279). Figure 1 lists four outputs in order: "The Case as Analytical Chronology," "The Diagnostic Case," "The Interpretative/Theoretical Case(s)," and "Meta Level Analysis across Cases" (p. 280).

The analytical chronology. The first output is the case as analytical chronology: "The requirement here is to lay out the narrative, to tell the story, but to tell the story across levels of analysis" (p. 280). His example interweaves the story of Jaguar Cars between 1960 and 1988 with the development of the world auto industry and the changing UK economy and politics, with "an explicit attempt to interpret and explain" the firm's performance through its inner and outer context (p. 280). Drawing on Strauss, he contrasts a case history, "characterized by temporal presentation," with a case study, which features "analytic abstractions for purposes of presenting theory"; an analytical chronology "is more like Strauss's (1987) case study than his case history" (p. 280). Such chronologies "are a psychological marker of progress and a necessary prelude to the cross case analysis and more explicit pattern formation which follows" (p. 280).

Later outputs and audience. Later outputs move beyond the chronology: interpretative cases link "emerging conceptual and theoretical ideas inductively derived from the case" to wider theoretical debates (p. 280), and cross-case work is organized "entirely around major concepts and themes," with cases used only to illustrate (p. 281). Chronologies are internal working documents, "written for ourselves as part of the process of sharing, learning, and pattern recognition" (p. 281). The section closes on why this structure matters: without it, researchers face "death by data asphyxiation" (p. 281).

Step 5 Compare cases for second order themes

A. Qualitative best practices *(from the master document, revised)*

- **Constant comparison** | Compare coded material against each other, explicitly, for similarities and differences. | "*Constant comparisons: The analytic process of comparing different pieces of data against each other for similarities and differences*" (Corbin & Strauss, 2015, p. 85)
- **Counter processing biases** | Look at the data in several divergent ways, so that volume, vivid cases or first impressions don't drive the themes. | "*Thus, the key to good cross-case comparison is counteracting these tendencies by looking at the data in many divergent ways.*" (Eisenhardt, 1989, p. 540)
- **Sort on dimensions** | Choose categories or dimensions that make sense for the research question and for what the codes and timelines show. Sort all cases on one of them, and compare the similarities among cases in the same group with the differences between groups; then re-sort on a second, then a third, and so on. Keep the dimensions that reveal meaningful patterns, and record the ones that don't. | "*One tactic is to select categories or dimensions, and then to look for within-group similarities coupled with intergroup differences.*" (Eisenhardt, 1989, p. 540)
- **Forced pair comparisons** | Compare cases that look similar to find differences, and cases that look different to find similarities. | "*The juxtaposition of seemingly similar cases by a researcher looking for differences can break simplistic frames.*" (Eisenhardt, 1989, p. 541)
- **Step up to second-order themes** | Move from informants' terms to concepts that describe and explain. Record which codes feed each theme, but don't link themes into a model yet. | "*In this 2nd-order analysis, we are now firmly in the theoretical realm, asking whether the emerging themes suggest concepts that might help us describe and explain the phenomena we are observing.*" (Gioia et al., 2013, p. 20) Also see (Gioia et al., 2013, p. 21).
- **Coherent within, distinct between** | Test each theme: the material inside it should hang together, and themes should differ clearly from one another. | "*Categories should then be judged by two criteria: internal homogeneity and external heterogeneity.*" (Patton, 2002, p. 465)
- **Size themes to the data** | Each second-order theme should draw on at least two first-order codes. One published example has 2–4 per theme, but the right number depends on the codes, the context and the research question, and can be much higher. Keep themes specific enough that they can still be grouped into broader categories at the next step. | "*Once a workable set of themes and concepts is in hand (and the culmination of the theme and concept development process leads to what Glaser and Strauss [1967] termed "theoretical saturation"), we investigate whether it is possible to distill the emergent 2nd-order themes even further into 2nd-order "aggregate dimensions."*" (Gioia et al., 2013, p. 20) Also see (Gioia et al., 2013, p. 21, Figure 1).

B. Further reading

B1. Eisenhardt, K. M. (1989). Building theories from case study research. Academy of Management Review, 14(4). "Searching for Cross-Case Patterns," pp. 540–541.

Eisenhardt starts from a warning about human judgment: "people are notoriously poor processors of information. They leap to conclusions based on limited data …, they are overly

influenced by the vividness … or by more elite respondents …, they ignore basic statistical properties …, or they sometimes inadvertently drop disconfirming evidence" (p. 540). The risk is "premature and even false conclusions" (p. 540). Cross-case search rests on prior within-case analysis, whose purpose is "to become intimately familiar with each case as a stand-alone entity," so that "the unique patterns of each case" emerge before generalizing (p. 540).

Tactic 1: sort on dimensions. The first tactic is to "select categories or dimensions, and then to look for within-group similarities coupled with intergroup differences" (p. 540). Here a "group" is a set of cases that land together when sorted on a dimension, not a team, so the comparison runs among cases in the same group and between groups of cases. This is different from within-case analysis, the stand-alone study of each case that comes first. Dimensions "can be suggested by the research problem or by existing literature, or the researcher can simply choose some dimensions" (p. 540). In one study, cases were sorted by founder-run versus professional management, high versus low performance, first- versus second-generation product, and large versus small size: "Some categories such as size and product generation revealed no clear patterns, but others such as performance led to important patterns" (p. 540). The tactic extends to a 2 × 2 or other cell design, or to a continuous scale that can be graphed.

Tactic 2: forced pairs. The second is to "select pairs of cases and then to list the similarities and differences between each pair" (p. 540). In one study, CEO power differences dominated first impressions, but paired comparison showed "that the speed of the decision process was equally important" (p. 541). Pairs can be extended to groups of three or four.

Tactic 3: split by data source. The third is to "divide the data by data source": one researcher combs observations, another interviews, another questionnaire evidence (p. 541). "When a pattern from one data source is corroborated by the evidence from another, the finding is stronger and better grounded." When evidence conflicts, deeper probing may reconcile it, or the conflict may expose "a spurious or random pattern, or biased thinking in the analysis" (p. 541). A variation splits the cases into groups and analyzes one group before the other.

The common aim is "to force investigators to go beyond initial impressions, especially through the use of structured and diverse lenses on the data," which makes accurate theory more likely and helps capture "novel findings" (p. 541).

B2. Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research. Organizational Research Methods, 16(1). "The analyses," pp. 20–22.

Gioia and colleagues describe how informant-level codes become theory-level themes. Early on, "A myriad of informant terms, codes, and categories emerge": "There could easily be 50 to 100 1st-order categories that emerge from the first 10 interviews," and feeling lost is normal: "You gotta get lost before you can get found" (p. 20).

From first-order to second-order. The analysts then seek "similarities and differences among the many categories," which "eventually reduces the germane categories to a more manageable number (e.g., 25 or 30)" (p. 20). They label these with "phrasal descriptors (preferably retaining informant terms)" and ask, "Is there some deeper structure in this array?", thinking at two levels at once: informant terms and "the more abstract, 2nd-order theoretical level of themes, dimensions, and the larger narrative" (p. 20). They focus on "nascent concepts that don't seem to have adequate theoretical referents in the existing literature" and on existing concepts that "leap out" because of their relevance to a new domain (p. 20). Once themes reach saturation, they consider distilling them into "aggregate dimensions" (p. 20).

The data structure. Terms, themes and dimensions are assembled into "a data structure (see Figure 1)—perhaps the pivotal step in our entire research approach," which gives "a graphic representation of how we progressed from raw data to terms and themes" (p. 20). It deliberately leaves out relationships: "this does not mean, however, that the data structure should capture relationships among the 2nd-order themes (a step that comes later in the theorizing process)" (p. 21). Their mantra is "No data structure; know nothing" (p. 21). The Figure 1 example (from Corley & Gioia, 2004) arranges 23 first-order concepts into 8 second-order themes and 3 aggregate dimensions, with 2–4 first-order concepts per theme (p. 21).

Disagreements and safeguards. "If agreements about some codings are low, we revisit the data, engage in mutual discussions, and develop understandings for arriving at consensual interpretations" (p. 22). Intercoder agreement statistics are optional: "we do so simply as another way to bolster our own confidence" (p. 22). Earlier in the article they add a guard against "going native": "we always have one member of the collaborative team adopt an outsider perspective—a devil's advocate, really, whose role it is to critique interpretations that might look a little too gullible" (p. 19).

Step 6 Develop categories and classify cases

A. Qualitative best practices *(from the master document, revised)*

- **Sharpen constructs** | Define each category precisely, then build the evidence for it case by case. | "*One step in shaping hypotheses is the sharpening of constructs. This is a two-part process involving (1) refining the definition of the construct and (2) building evidence which measures the construct in each case.*" (Eisenhardt, 1989, p. 541)
- **Cross the dimensions** | Cross the underlying dimensions into a matrix and check every cell against the data, including cells that turn out empty. | "*The logical process involves creating potential categories by crossing one dimension or typology with another, and then working back and forth between the data and one's logical constructions, filling in the resulting matrix.*" (Patton, 2002, p. 468) Also see (Patton, 2002, p. 470).
- **Keep categories processual** | Don't collapse a case's trajectory into one static score. Categories can be defined by change over time. | "*However, this is not process theory; the complexities of the probabilistic interaction of events, parallel and alternate tracks, patterns of mutual shaping over time, and evolving performance have been compressed into positions on a small number of scales ...*" (Langley, 1999, p. 704)
- **Replication logic** | Test the categories one case at a time; each case either confirms or disconfirms the scheme. | "*The underlying logic is replication, that is, the logic of treating a series of cases as a series of experiments with each case serving to confirm or disconfirm the hypotheses.*" (Eisenhardt, 1989, p. 542)
- **Seek negative cases** | Examine cases that fit no category or fit several; many misfits mean the scheme needs revising. | "*Where patterns and trends have been identified, our understanding of those patterns and trends is increased by considering the instances and cases that do not fit within the pattern.*" (Patton, 2002, p. 554) Also see (Patton, 2002, p. 465).
- **Weigh misfits by significance, not count** | A case that differs from a category only in detail needn't trigger revision; record it. A case that contradicts a category's core idea must be examined, even if it is the only one, because disconfirming cases often reveal a boundary condition. Many minor misfits together also mean the scheme needs revising. | "*Cases which disconfirm the relationships often can provide an opportunity to refine and extend the theory.*" (Eisenhardt, 1989, p. 542) Also see (Patton, 2002, p. 465).

B. Further reading

B1. Patton, M. Q. (2002). Qualitative Research and Evaluation Methods (3rd ed.). "Analyst-Constructed Typologies," pp. 458–462, and "Logical Analysis," pp. 468–471.

Typologies, indigenous first. Patton defines typologies as "classification systems made up of categories that divide some aspect of the world into parts along a continuum." Unlike taxonomies, which classify "through mutually exclusive and exhaustive categories," typologies "are built on ideal-types or illustrative endpoints" (p. 457). He starts with indigenous typologies, the distinctions members draw themselves, such as teachers who labeled truant students "chronics" or "borderlines" (p. 458). Analyst-constructed typologies come after: "Once indigenous concepts, typologies, and themes have been surfaced, the analyst moves to a different task of induction" (pp. 458–459).

Care and testing. "Such constructions must be done with considerable care to avoid creating things that are not really in the data" (p. 459). The analyst identifies "patterns that appear to exist but remain unperceived by the people studied," and "The danger is that analyst-constructed typologies impose a world of meaning on the participants that better reflects the observer's world than the world under study" (pp. 459–460). The test, quoting Lofland: "The best and most stringent test of observer constructions is their recognizability to the participants themselves" (p. 460).

Examples. A museum evaluation sorted visitors into four progressively more engaged types, "The Commuter," "The Nomad," "The Cafeteria Type," and "The V.I.P.—Very Interested Person," and the typology then shaped how interview responses were weighed (pp. 460–461). A study of homelessness produced four types of homeless people plus three categories of time use ("Hanging out," "Getting by," "Getting ahead") (p. 462). The lesson: "the first purpose of typologies is to distinguish aspects of an observed pattern or phenomenon descriptively"; interpretation comes later (p. 462).

Logical analysis. Once some dimensions exist, "it is sometimes useful to cross-classify different dimensions to generate new insights" (p. 468). In a dropout program, crossing teachers' preferred intervention (rehabilitation, maintenance, punishment) with their behavior (taking responsibility versus shifting it to others) produced six roles, from "Counselor/Friend" to "Ostrich" (Exhibit 8.8, pp. 468–469). The analyst, who had struggled with the data, said "the whole thing immediately fell into place" (p. 468). Patton warns: "It is easy for a matrix to begin to manipulate the data as the analyst is tempted to force data into categories created by the cross-classification to fill out the matrix and make it work" (pp. 469–470). Logically derived categories "must be tested out and confirmed by the actual data" (p. 470). Empty cells call for abductive exploration: each points to a category "that either has been overlooked in the data or is logically a possibility in the setting but has not been manifested," and the latter "are important to note because their importance derives from the fact that they did not occur" (p. 470).

B2. Eisenhardt, K. M. (1989). Building theories from case study research. Academy of Management Review, 14(4). "Shaping Hypotheses," pp. 541–544.

Once tentative themes and concepts emerge, the next step is to "compare systematically the emergent frame with the evidence from each case in order to assess how well or poorly it fits with case data" (p. 541). "The central idea is that researchers constantly compare theory and data—iterating toward a theory which closely fits the data" (p. 541).

Sharpening constructs. Constructs are sharpened through "constant comparison between data and constructs so that accumulating evidence from diverse sources converges on a single, well-defined construct" (p. 541). Sutton and Callahan, for example, built constructs such as "disengagement" and "denigration via rumor" from telephone conversations, interviews, court records, observation of a creditors' meeting and press coverage (pp. 541–542). This resembles establishing construct validity, except that "the construct, its definition, and measurement often emerge from the analysis process itself" (p. 542). Because indicators vary across cases, "many researchers rely on tables which summarize and tabulate the evidence underlying the construct" (p. 542). Table 3 shows the format: one row per firm, with columns for a description of the CEO, power scores, CEO-dominated functions, decision style, and story examples with the speaker's role in parentheses (p. 543).

Testing relationships case by case. The second step is "verifying that the emergent relationships between constructs fit with the evidence in each case. Sometimes a relationship is

confirmed by the case evidence, while other times it is revised, disconfirmed, or thrown out for insufficient evidence" (p. 542). Confirming cases "enhance confidence in the validity of the relationships," while disconfirming cases "often can provide an opportunity to refine and extend the theory" (p. 542). In one study, a case contradicted the idea that political coalitions are stable; it turned out the team was newly formed, and the theory was refined to "indicate that increasing stabilization of coalitions occurs over time" (p. 542). Qualitative data then supply the "why" behind each relationship (p. 542).

Judgment and display. Without statistical tests, this work is "more judgmental": "The research team must judge the strength and consistency of relationships within and across cases and also fully display the evidence and procedures when the findings are published, so that readers may apply their own standards" (p. 544).

Step 7 Iterate between data coding theory and literature

A. Qualitative best practices *(from the master document, revised)*

- **Literature sensitizes, not dictates** | Use prior concepts to suggest where to look, then check how they show up, if at all, in these cases. | "*Sensitizing concepts give the analyst "a general sense of reference" and provide "directions along which to look" (Blumer 1969: 148).*" (Patton, 2002, p. 456) Also see (Gioia et al., 2013, p. 21).
- **Compare with similar and conflicting literature** | For each category, ask what it resembles, what it contradicts and why, drawing on a broad range of work. | "*This involves asking what is this similar to, what does it contradict, and why. A key to this process is to consider a broad range of literature.*" (Eisenhardt, 1989, p. 544)
- **Return to the evidence in each case** | After each revision, re-check the categories and assignments against every case's data. | "*A second step in shaping hypotheses is verifying that the emergent relationships between constructs fit with the evidence in each case.*" (Eisenhardt, 1989, p. 542)
- **Stop at saturation** | Stop iterating when another round barely improves the categories, not after a fixed number of rounds. Treat a round as minor when it produces no new, split, merged or dropped categories, only wording changes or a few case reassignments. | "*That is, the iteration process stops when the incremental improvement to theory is minimal.*" (Eisenhardt, 1989, p. 545)

B. Further reading

B1. Eisenhardt, K. M. (1989). Building theories from case study research. Academy of Management Review, 14(4). "Enfolding Literature" and "Reaching Closure," pp. 544–545.

Conflicting literature. "An essential feature of theory building is comparison of the emergent concepts, theory, or hypotheses with the extant literature" (p. 544). Conflicting literature matters for two reasons: "if researchers ignore conflicting findings, then confidence in the findings is reduced," and "conflicting literature represents an opportunity," forcing "a more creative, framebreaking mode of thinking" (p. 544). In one study, politics turned up under centralized power, although prior work tied politics to decentralized power. Probing both showed that both extremes create frustration that leads to politics, a reconciliation that "integrated the conflicting findings into a single theoretical perspective" (p. 544).

Similar literature. Literature with similar findings "ties together underlying similarities in phenomena normally not associated with each other," producing "a theory with stronger internal validity, wider generalizability, and higher conceptual level" (p. 544). Leonard-Barton linked her findings on technology and organizations adapting to each other to similar findings in the education literature; Gersick linked a midpoint transition in project groups to punctuated equilibrium and the adult midlife transition (pp. 544–545). Tying results to literature is "particularly crucial in theory-building research because the findings often rest on a very limited number of cases" (p. 545).

Reaching closure. Two decisions close the analysis: when to stop adding cases and when to stop iterating. For cases, the criterion is theoretical saturation, "simply the point at which incremental learning is minimal because the researchers are observing phenomena seen before," and "while there is no ideal number of cases, a number between 4 and 10 cases usually works well" (p. 545). For iteration, "saturation is the key idea. That is, the iteration process

stops when the incremental improvement to theory is minimal" (p. 545), much like ending the revision of a manuscript. The result may be concepts, a conceptual framework or propositions, and it may disappoint: "The research may simply replicate prior theory, or there may be no clear patterns within the data" (p. 545).

B2. Patton, M. Q. (2002). Qualitative Research and Evaluation Methods (3rd ed.). "Sensitizing Concepts," pp. 456–457.

Sensitizing concepts are "categories that the analyst brings to the data," with origins "in social science theory, the research literature, or evaluation issues identified at the beginning of a study" (p. 456). Quoting Blumer, they give the analyst "a general sense of reference" and provide "directions along which to look"; using them "involves examining how the concept is manifest and given meaning in a particular setting or among a particular group of people" (p. 456). A concept can be both indigenous and sensitizing: in a leadership program, "leadership" was indigenous because staff and participants used it constantly, and sensitizing because the evaluators knew going in that it would matter (p. 456).

An example and its rules. Conroy applied the concept of "victimization" to police officers, turning the usual idea of the victim around, and found it helpful in understanding "the isolation, lack of interpersonal affect, cynicism, repressed anger, and sadness" he observed (p. 456). He presented the officers' own words so that "readers are able to make their own determination of whether the concept 'victimization' helps in making sense of the data" (p. 457). Patton draws three rules. "The analyst's constructs should not dominate the analysis, but rather should facilitate the reader's understanding of the world under study." "Concepts are never a substitute for direct experience with the descriptive data." And "The reader can usually tell when the analyst is more interested in proving the applicability and validity of a concept than in letting the data reveal the perspectives of the people interviewed" (p. 457).

Step 8 Build the findings narrative and evidence display

A. Qualitative best practices *(from the master document, revised)*

- **Organize by the theory** | Structure the narrative around the model's parts, each backed by case evidence. | "*The best way to address this challenge of "better stories vs. better theories" is to develop a theory in sections or by distinct propositions in such a way that each is supported by empirical evidence.*" (Eisenhardt & Graebner, 2007, p. 29)
- **Construct tables** | Use a table to show the evidence for each construct across many cases, complementing the exemplar narratives. | "*A separate table that summarizes the evidence for each theoretical construct is a particularly effective way to present the case evidence.*" (Eisenhardt & Graebner, 2007, p. 29)
- **Enough quotation to enter the situation** | Include enough description and direct quotes that readers hear the members' voices, without trivia. | "*Sufficient description and direct quotations should be included to allow the reader to enter into the situation and thoughts of the people represented in the report.*" (Patton, 2002, p. 503) Also see (Gioia et al., 2013, p. 17).
- **Show multiple data sources** | Let findings display evidence from each source type collected, not only interviews. | "*We want to encourage scholars to provide evidence of data triangulation by showing, in their findings sections, examples from the multiple data sources they describe in their methods sections, thereby providing a wider window for their data "to speak."*" (Rouse & Harrison, 2015, p. 286)
- **Report the misfits** | Present cases that don't fit the main pattern and alternative explanations, not just the best examples. | "*But I would note that the section of the report that involves exploration of alternative explanations and consideration of why certain cases do not fall into the main pattern can be among the most interesting sections of a report to read.*" (Patton, 2002, p. 555)
- **Exemplars in text, full evidence in tables** | For each construct, tell a few exemplar cases as short stories and include a few exemplar quotes; put the evidence from every other case in construct tables. How much appears in the text depends on the number of constructs and the length of the article. | "*In other words, the use of summary tables and aids that summarize the case evidence complements the selective story descriptions of the text and further emphasizes the rigor and depth of the empirical grounding of the theory.*" (Eisenhardt & Graebner, 2007, p. 29)

B. Further reading

B1. Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. Academy of Management Journal, 50(1). "Presenting Empirical Evidence," pp. 28–29, with "Writing the Emergent Theory," pp. 29–30.

The problem. Deductive studies summarize evidence in compact numerical tables, "But case data cannot be so tightly summarized, because much of it consists of rich qualitative detail" (p. 29). A single-case study can present "a relatively complete rendering of the story," intertwined with the theory (p. 29). Multiple-case research can't: "If the researcher relates the narrative of each case, then the theory is lost and the text balloons" (p. 29).

The solution. Develop the theory in sections or propositions, each backed by evidence: "Thus, the overarching organizing frame of the paper is the theory, and each part of the theory is

demonstrated by evidence from at least some of the cases" (p. 29). Because not every proposition can be supported by every case in the text, "extensive tables and other visual devices that summarize the related case evidence are central to signaling the depth and detail of empirical grounding" (p. 29). Construct tables "summarize the case evidence and indicate how the focal construct is 'measured,' thus increasing the 'testability' of the theory" (p. 29). Readers expecting rich narrative may be disappointed, so it is "critical to invest in developing well-crafted tables, appendixes, and visual aids … and to tie those tables clearly to the text" (p. 29). Telling the story first and the theory second is "reasonable and common," but it "moves the theory off center stage" (p. 29, n. 1).

Writing the theory several ways. The authors recommend writing the theory "in multiple ways": sketch it in the introduction; in the body, state each proposition "and link it to the supporting empirical evidence for each construct and for the proposed relationship between the constructs"; write "the underlying theoretical arguments that provide the logical link between the constructs," drawn "from case evidence (e.g., an informant explaining the logic) and/or from more detached logic"; and provide "a visual theory summary such as a 'boxes and arrows' diagram or summary table" (pp. 29–30). When the work is well done, "the propositions will be consistent with most (or even all) of the cases" (p. 29).

B2. Patton, M. Q. (2002). Qualitative Research and Evaluation Methods (3rd ed.). "Balance Between Description and Interpretation," pp. 503–504.

How much description. "Description and quotation provide the foundation of qualitative reporting" (p. 503), but there are limits on both sides. "Description should stop short, however, of becoming trivial and mundane. The reader does not have to know everything that was done or said. Focus comes from having determined what's substantively significant and providing enough detail and evidence to illuminate and make that case" (p. 503). Equally, "the description must not be so 'thin' as to remove context or meaning" (p. 503). Quoting Denzin, thick description "presents detail, context, emotion, and the webs of social relationships that join persons to one another," so that "the voices, feelings, actions, and meanings of interacting individuals are heard" (p. 503).

Balancing description with interpretation. "Endless description becomes its own muddle. The purpose of analysis is to organize the description so that it is manageable" (p. 503). A readable report "provides sufficient description to allow the reader to understand the basis for an interpretation, and sufficient interpretation to allow the reader to appreciate the description" (pp. 503–504). Brief notes in the text can show how a finding was checked, such as that participants agreed with the analysis or that "this connection was independently arrived at by two analysts looking at the data separately" (p. 504). The analyst should also signal how strong each finding is: "When are patterns 'clear'? When are they 'strongly supported by the data'? When are the patterns 'merely suggestive'?" (p. 504).

Step 9 Construct the process model

A. Qualitative best practices *(from the master document, revised)*

- **Focus on the arrows** | Building the model means specifying how the concepts relate dynamically, not just listing them. | "*The key question for us as model builders is how to account for not only all the major emergent concepts, themes, and dimensions, but also for their dynamic interrelationships.*" (Gioia et al., 2013, p. 22)
- **Model mirrors the data structure** | Every box should trace to the Step 5–7 data structure, and the model should add only the relationships among those boxes. | "*A reader should be able to look at the grounded theory model and see that the essential concepts, themes, and/or dimensions contained in the data structure are well represented in the model, but that the relational dynamics among those concepts are now made transparent.*" (Gioia et al., 2013, p. 22)
- **Explain why each link holds** | For each link, state the mechanism and point to the case evidence for it. | "*Therefore, it is important to discover the underlying theoretical reasons for why the relationship exists.*" (Eisenhardt, 1989, p. 542) Also see (Eisenhardt & Graebner, 2007, p. 29).
- **Members' causal stories vs. analyst inference** | Keep apart links that informants themselves describe and links the analyst infers. | "*To the extent that you are describing the causal linkages suggested by and believed in by those you've interviewed, you haven't crossed the line from description into causal interpretation.*" (Patton, 2002, p. 478)
- **Test rival models** | Build alternative ways of linking the concepts, look for data that support them, and report what was considered. | "*Keep track of and report alternative classification systems, themes, and explanations that you considered and "tested" during data analysis.*" (Patton, 2002, p. 553)
- **Keep links that replicate** | Keep a link if it holds in all or nearly all cases; with 10 or more cases, roughly 1 in 10 may lack evidence for it. A case that contradicts a link, rather than simply lacking evidence, must be explained; an unexplained contradiction means revising or dropping the link. | "*In contrast, multiple-case researchers retain only the relationships that are replicated across most or all of the cases.*" (Eisenhardt & Graebner, 2007, p. 30)

B. Further reading

B1. Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research. Organizational Research Methods, 16(1). "From Data Structure to Grounded Theory," p. 22, and Figure 2, p. 23.

From photograph to motion picture. However much work goes into it, the data structure "is nonetheless a static picture of a dynamic phenomenon, and process research doesn't actually investigate processes unless the static picture—a photograph, if you will—can be made into a motion picture" (p. 22). The goal is a grounded theory model "that shows the dynamic relationships among the emergent concepts that describe or explain the phenomenon of interest and one that makes clear all relevant data-to-theory connections" (p. 22).

The arrows. In boxes-and-arrows terms, "this process amounts to assembling the constellation of boxes with a special focus on the arrows. It is the arrows that 'set everything in motion'" (p. 22). Considering how the concepts relate makes possible "theoretical insights that would not be apparent simply by inspecting the static data structure itself," and "there is room for a

conceptual leap in this process as well," which the first author calls a "Shazzam!", grounded in close familiarity with the data and the data structure (p. 22).

The worked example. Figure 2 (from Corley & Gioia, 2004) runs left to right from "Pre-Spin-off Identity" to "Post-Spin-off Identity." Between them sit the three triggers of identity ambiguity; a central "Identity Change Context" containing identity ambiguity and the sensegiving imperative; and the three leaders' responses (p. 23). The elements come from the Figure 1 data structure; what the model adds is order and connection. The authors sum up the relationship: "If the data structure is the anatomy of the coming theory, then the grounded model is the physiology of that theory" (p. 24).

B2. Langley, A. (1999). Strategies for theorizing from process data. Academy of Management Review, 24(4). "Temporal Bracketing Strategy," pp. 703–704.

Periods, not phases. The time axis in Langley's Figure 2 is divided into three "periods," which "do not have any particular theoretical significance. They are not 'phases' in the sense of a predictable sequential process but, simply, a way of structuring the description of events." They reflect "a certain continuity in the activities within each period" and "certain discontinuities at its frontiers" (p. 703). Beyond description, periods become "comparative units of analysis for the exploration and replication of theoretical ideas," which is "especially useful if there is some likelihood that feedback mechanisms, mutual shaping, or multidirectional causality will be incorporated into the theorization" (p. 703).

The logic. Borrowing from structuration theory, Langley notes that mutual influences are hard to capture at the same time, so it is easier to analyze them in sequence. Splitting the data into successive periods "enables the explicit examination of how actions of one period lead to changes in the context that will affect action in subsequent periods" (p. 703). Barley's study of two radiology departments traced how the initial context shaped interactions, which then changed the context and started a new phase; Denis et al. traced management tactics and leadership roles "over five periods, separated by discontinuities in team membership," and observed alternating dynamics (p. 703). "With this strategy, a shapeless mass of process data is transformed into a series of more discrete but connected blocks." Within each block, the analyst describes the processes, examines how the context affects them and what they do to the future context, and "Discontinuities lead to replication of the analysis in a new phase" (p. 703).

Strengths and limits. The strategy handles eclectic data, and "one or a few cases may be sufficient to produce useful insights" (p. 704). But "there is no a priori guarantee that discontinuities will naturally synchronize themselves to produce unequivocal periods," so accuracy depends on "the appropriateness of the temporal decomposition and the robustness of the analysis to different periodizations" (p. 704). Langley contrasts this with compressing a process into a few variables, which "is not process theory" (p. 704).